

Becoming More Fluent with Function Notation and Reviewing Previous Math (Homework)

① If $f(x) = x - 7$, and $f(m) = -4$, find m .

② If $f(x) = 1 - \frac{2x}{5}$, and $f(p) = 5$, find p .

③ If $f(x) = -6x + \frac{1}{2}$, and $f(b) = 3$, find b .

④ If $f(x) = \frac{1}{3}(x - 4) + 2$, and $f(c) = -1$, find c .

⑤ If $f(x) = x^2 + 7x$, and $f(a) = -6$, find a .

⑥ If $f(x) = x^2 - 9x + 21$, and $f(d) = 3$, find d .

⑦ If $f(x) = 2x^2 - 3x - 18$, and $f(k) = -4$, find k .

⑧ If $f(x) = 21x^2 + 42x + 9$, and $f(r) = 9$, find r .

⑨ If $f(x) = 3x^2 - 10x + 1$,
and $f(v) = -2$, find v .

⑩ If $f(x) = \frac{(2 + \frac{x}{3})^2}{4} + 2$,
and $f(b) = 4$, find b .

⑪ If $f(x) = -5(3 - 9x)^2 + 2$, and $f(c) = -2$, find c .

⑫ If $f(x) = \frac{\sqrt{\frac{x}{5}-1}}{3} - 7$,
and $f(m) = -5$, find m .

⑬ If $f(x) = 3\sqrt{2x+5} - 8$,
and $f(n) = -7.5$, find n .

⑭ If $f(x) = \frac{1}{x+1}$,
and $f(p) = \frac{1}{3}$, find p .

⑮ If $f(x) = \frac{-2}{-3x+1}$,
and $f(a) = -1$, find a .

①⑥ If $f(x) = -5x + 3$, and $g(x) = 5x + 1$, for what value of x does $f(x) = g(x)$?

①⑦ If $f(x) = -6(x - 6) + 4$, and $h(x) = x - 4$, for what value of x is $f(x) = h(x)$?

①⑧ If $g(x) = 2x - 1$ and $h(x) = -x + 1$, find the coordinates of the point where the graphs of these functions intersect.

①⑨ If $g(x) = x^2 + 3x + 3$ and $h(x) = -x - 1$, find the coordinates of the point where the graphs of these functions intersect.

②⑩ If $g(x) = 3x^2 + 12x - 8$ and $f(x) = 5x^2 + 3x + 2$, for what value(s) of x does $g(x) = f(x)$?

②1) If $h(x) = -x^2 + 27x$ and $f(x) = 2x^2 - 10x + 90$, Find the coordinates of the point(s) where the graphs of these functions intersect.

②2) If $f(x) = 3(x-4)^2 + 1$, and $h(x) = 10$, for what value of x is $f(x) = h(x)$?

②3) If $f(x) = \frac{\sqrt{12x-4}}{8} - 10$, and $g(x) = 1$, find the coordinates of the point where the graphs of these functions intersect.

Homework Answers

① $m=3$

② $p=-10$

③ $b=-\frac{5}{12}$

④ $c=-5$

⑤ $a=-1 \text{ or } -6$

⑥ $d=3 \text{ or } 6$

⑦ $k=-2 \text{ or } \frac{7}{2}$

⑧ $v=0 \text{ or } -2$

⑨ $v=3 \text{ or } \frac{1}{3}$

⑩ $b=\pm 6\sqrt{2}-6$

⑪ $c=\frac{\pm 2\sqrt{5}+15}{45}$

⑫ $m=185$

⑬ $n=-\frac{179}{72}$

⑭ $p=2$

⑮ $a=-\frac{1}{3}$

⑯ $x=\frac{1}{5}$

⑰ $x=\frac{44}{7}$

⑱ $(\frac{2}{3}, \frac{1}{3})$

⑲ $(-2, 1)$

⑳ $x=2 \text{ or } \frac{5}{2}$

㉑ $(9, 162) \text{ and } (3\frac{1}{3}, 78\frac{8}{9})$

㉒ $x=\pm\sqrt{3}+4$

㉓ $(645\frac{2}{3}, 1)$